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Adjoint Functors and Universal Properties

Complete Overview

Adjoint functors are pairs of functors that formalize “optimal approximations” and “universal solutions” to mathematical problems. They are arguably the most important concept in category theory.

Definition

Adjoint Pair: $F \dashv G$

Functors $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$ form an **adjoint pair** (F left adjoint to G) when there exists a **natural bijection**:

$$\text{Hom}_{\mathcal{D}}(F(c), d) \cong \text{Hom}_{\mathcal{C}}(c, G(d))$$

Naturally in c and d : The isomorphism respects morphisms in both categories.

Three Equivalent Formulations

1. Hom-Set Adjunction (Definition Above)

- **Most direct:** Explicit bijection between hom-sets
- **Easiest to compute:** Work directly with morphisms

2. Unit-Counit Formulation

Given natural transformations: - **Unit:** $\eta: \text{id}_C \Rightarrow G \circ F$ - **Counit:** $\varepsilon: F \circ G \Rightarrow \text{id}_D$

Triangle identities:

$$F \rightarrow G \circ F \circ F \rightarrow G \circ F \quad (\text{Left triangle})$$

$$\text{id} \circ F = F$$

$$G \rightarrow F \circ G \circ G \rightarrow F \circ G \quad (\text{Right triangle})$$

$$G \circ \text{id} = G$$

3. Universal Morphism Formulation

For each $c \in C$, there exists universal morphism:

$$\eta_c: c \rightarrow G(F(c))$$

Such that for any $f: c \rightarrow G(d)$, there exists **unique** $g: F(c) \rightarrow d$ with:

$$f = G(g) \circ \eta_c$$

Why Adjoints Are the “Big Idea”

Unification Across Mathematics: - All free constructions (free groups, free modules, free rings) - All universal properties (products, coproducts, tensor products) - All limits and colimits - All Kan extensions - All representable functors

Quote from Mac Lane: > “The concept of adjoint functors is to category theory what the Fundamental Theorem is to Calculus.”

Concrete Examples

1. Free-Forgetful Adjunction

- **F: Set $\rightarrow$ Grp** (free group functor)
- **G: Grp $\rightarrow$ Set** (forgetful functor)
- **Adjunction:** $F \dashv G$

For any set X and group H :

$$\text{Hom}_{\text{Grp}}(F(X), H) \cong \text{Hom}_{\text{Set}}(X, U(H))$$

Left side: group homomorphisms from free group on X Right side: set functions from X into underlying set of H

2. Tensor-Hom Adjunction

- **$**F = M \otimes_R -$** : Tensor product with fixed module

- **$G = \text{Hom}_R(M, -)$** : Hom functor
- **Adjunction**: $F \dashv G$

$$\text{Hom}_R(M \otimes_R N, P) \cong \text{Hom}_R(N, \text{Hom}_R(M, P))$$

Bilinear maps $\rightarrow$ R-linear maps (currying)

3. Diagonal-Limit Adjunction

- **$\Delta: C \rightarrow C^J$** (diagonal functor, constant functor)
- **$\lim: C^J \rightarrow C$** (limit)
- **Adjunction**: $\lim \dashv \Delta$

$$\text{Hom}_{C^J}(F, \Delta(c)) \cong \text{Hom}_C(\lim F, c)$$

Natural cones to constant $\rightarrow$ morphisms from limit

Key Preservation Property: RAPL

Right Adjoint Preserves Limits

Theorem: If $F \dashv G$ and $G: D \rightarrow C$, then **G preserves all limits** that exist in D .

Dual: Left adjoints preserve colimits.

Proof Sketch: Using hom-set adjunction with limit's universal property.

Examples: - Forgetful functors preserve limits (right adjoints) - Free functors reflect limits (left adjoints) - Hom functors preserve limits (right adjoints)

General and Special Adjoint Functor Theorems

General Adjoint Functor Theorem

If $G: D \rightarrow C$ and: 1. C is locally small 2. D is locally small 3. G satisfies solution set condition

Then $F \dashv G$ exists.

Special Case (Freyd)

If $G: D \rightarrow C$ and: 1. G preserves limits 2. For each $c \in C$, the comma category $(c \downarrow G)$ has small initial object

Then $F \dashv G$ exists.

Beck Monadicity Theorem

Theorem: A right adjoint functor $G: D \rightarrow C$ is **monadic** (factorizes through category of algebras over monad $T = G \circ F$) if and only if: 1. G reflects isomorphisms 2. G preserves coequalizers of G -split pairs 3. Coequalizers exist in D

Consequence: Algebraic categories are exactly monadic over Set .

Locally Presentable Categories

Definition: Category $\mathcal{C}$ is κ -**presentable** if: - It has κ -filtered colimits - $G: [\mathcal{C}^{\text{op}}, \text{Set}] \rightarrow \text{Set}$ (Yoneda embedding) has left adjoint

Theorem: Locally presentable categories admit enough adjoints and are “algebraic” in nature.

Examples: - Set , Grp , Mod_R , Top (all locally presentable) - Presheaf categories $[\mathcal{C}^{\text{op}}, \text{Set}]$

Why Adjoints Preserve Structure

Universal Property: Adjoints characterize objects by their relationships (morphisms).

Preservation: Since adjoints respect morphism structure, they preserve: - Isomorphisms (always) - Limits (for right adjoints) - Colimits (for left adjoints) - Exactness (for certain adjoint pairs)

Applications

Algebra

- Free groups, rings, modules as left adjoints
- Tensor products as left adjoints
- Hom functors as right adjoints

Topology

- Fundamental group: $\text{Top} \rightarrow \text{Grp}$ (left adjoint to classifying space)
- Compactification: $\text{Top} \rightarrow \text{CompHaus}$ (left adjoint to inclusion)

Logic

- Free boolean algebras (left adjoint to forgetful)
- Stone duality: $\text{BA} \rightleftarrows \text{TopologicalSpaces}$ (adjoint pair)

Common Mistakes

✗ Mistake: “Adjoints are rare and exotic” **✓ Reality:** Most natural constructions have adjoints

✗ Mistake: “Left adjoint = right adjoint” **✓ Reality:** They have opposite preservation properties

✗ Mistake: “Adjoint is symmetric relation” **✓ Reality:** “ $F \dashv G$ ” reads as “ F is left adjoint TO G ”

Study Guide

1. Understand unit-counit formulation deeply
2. Work through free-forgetful examples
3. Prove RAPL theorem
4. Recognize adjoints in your field

5. Apply to universal problems

Further Reading

- Awodey, “Category Theory” (Chapter 9)
- Mac Lane, “Categories for the Working Mathematician” (Chapter IV)
- Leinster, “Basic Category Theory” (Chapter 6)
- Borceux, “Handbook of Categorical Algebra” (Vol 1, Chapter 3)

Key Questions

1. Why is the natural bijection critical to adjunction?
2. How do free-forgetful pairs exemplify adjunction?
3. Why do right adjoints preserve limits?
4. How are adjunctions related to Kan extensions?
5. What makes Beck monadicity theorem profound?